

An Elementary Obstruction to Uniform Prime-Gap Asymptotics and the Additive Structure of the Admissible Constants

Draft note

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Abstract

Let p_n be the n -th prime, let $g_n = p_{n+1} - p_n$, and put

$$d(x) = \max_{p_n < x} g_n.$$

For a fixed constant C , consider the uniform asymptotic

$$\pi(y + Cd(x)) - \pi(y) \sim \frac{Cd(x)}{\log y} \quad (x/2 < y < x).$$

We record the elementary obstruction coming from record prime gaps, prove that no $0 < C \leq 1$ can satisfy the asymptotic, and then use the difference observation that constants satisfying the asymptotic are closed under subtraction. It follows that the set of admissible constants is either trivial or a cyclic lattice $C_0\mathbb{Z}$, generated by a unique constant $C_0 > 1$.

1 The uniform assertion

Let p_n denote the n -th prime and write

$$g_n = p_{n+1} - p_n$$

for the corresponding prime gap. For $x > 2$, define

$$d(x) = \max_{p_n < x} (p_{n+1} - p_n).$$

Thus a gap is counted as soon as its left endpoint is below x ; the right endpoint of that gap need not be below x .

For $C \neq 0$, let $A(C)$ denote the assertion

$$\sup_{x/2 < y < x} \left| \frac{\pi(y + Cd(x)) - \pi(y)}{Cd(x)/\log y} - 1 \right| \longrightarrow 0 \quad (x \rightarrow \infty). \quad (1)$$

For the structural statement, we adjoin 0 as a formal trivial constant and set

$$\mathcal{C} = \{0\} \cup \{C \neq 0 : A(C) \text{ holds}\}.$$

The ratio in (1) is not meant to define $A(0)$; 0 is included only as the identity element for the additive-structure statement.

We use two standard facts. First, prime gaps are unbounded, hence strict record gaps occur infinitely often. Second, by the prime number theorem,

$$d(x) = o(x). \quad (2)$$

Indeed $p_{n+1}/p_n \rightarrow 1$, so every prime gap with left endpoint below x is $o(x)$.

2 The original obstruction

Theorem 1. *Let*

$$1 < C_1 < C_2, \quad 0 < C_2 - C_1 < \frac{1}{2}.$$

Then $A(C_1)$ and $A(C_2)$ cannot both hold.

Proof. Put $\eta = C_2 - C_1$, so $0 < \eta < 1/2$. Choose infinitely many strict record prime gaps

$$(P_k, Q_k) = (p_{n_k}, p_{n_k+1}), \quad D_k = Q_k - P_k,$$

so that $D_k > g_m$ for all $m < n_k$.

Choose

$$\alpha \in \left(\frac{1}{2} + \eta, 1\right)$$

and define

$$x_k = P_k + \alpha D_k, \quad y_k = P_k + \frac{1}{2} D_k, \quad z_k = P_k + \left(\frac{1}{2} + \eta\right) D_k.$$

Then

$$P_k < y_k < z_k < x_k < Q_k$$

and also $x_k/2 < y_k < z_k < x_k$. Since there are no primes in (P_k, Q_k) , the definition of $d(x)$ gives

$$d(x_k) = D_k.$$

Moreover $\pi(y_k) = \pi(z_k) = \pi(P_k)$.

Now

$$y_k + C_2 D_k = P_k + \left(\frac{1}{2} + C_2\right) D_k = z_k + C_1 D_k.$$

Let this common endpoint be R_k . Then

$$\pi(y_k + C_2 D_k) - \pi(y_k) = \pi(R_k) - \pi(y_k) = \pi(R_k) - \pi(z_k) = \pi(z_k + C_1 D_k) - \pi(z_k).$$

Call this common quantity N_k .

If both $A(C_2)$ and $A(C_1)$ held, then

$$N_k \sim \frac{C_2 D_k}{\log y_k} \quad \text{and} \quad N_k \sim \frac{C_1 D_k}{\log z_k}.$$

Consequently

$$\frac{C_2 \log z_k}{C_1 \log y_k} \longrightarrow 1.$$

But

$$1 < \frac{z_k}{y_k} = 1 + \frac{\eta D_k}{P_k + D_k/2} < 1 + 2\eta < 2,$$

so $\log z_k = \log y_k + O(1)$, and hence

$$\frac{\log z_k}{\log y_k} \longrightarrow 1.$$

The preceding displayed limit would therefore force $C_2/C_1 = 1$, impossible. □

Corollary 1. *The assertion $A(C)$ cannot hold for every fixed $C > 1$.*

3 No constants in $(0, 1]$

Theorem 2. *No constant $C \in (0, 1]$ satisfies $A(C)$.*

Proof. Again let (P_k, Q_k) be strict record prime gaps and set $D_k = Q_k - P_k$. Choose $x_k = P_k + \alpha D_k$, with fixed $0 < \alpha < 1$, in such a way that the starting points below lie in $(x_k/2, x_k)$. This is possible for all large k , since $D_k = o(P_k)$. For all such k , one has $d(x_k) = D_k$.

First suppose $0 < C < 1$. Put

$$y_k = P_k + \frac{1-C}{2} D_k.$$

Then both endpoints y_k and $y_k + CD_k = P_k + \frac{1+C}{2} D_k$ lie strictly inside the prime-free interval (P_k, Q_k) . Hence

$$\pi(y_k + CD_k) - \pi(y_k) = 0,$$

whereas the predicted main term $CD_k/\log y_k$ is positive. Thus the normalized ratio is 0, not 1, contradicting $A(C)$.

It remains to treat $C = 1$. Take $y_k = P_k$. Then

$$\pi(P_k + D_k) - \pi(P_k) = \pi(Q_k) - \pi(P_k) = 1.$$

If $A(1)$ held, this would imply

$$\frac{D_k}{\log P_k} \longrightarrow 1. \tag{3}$$

Let P_k^- be the prime immediately preceding P_k , and write

$$\Delta_k = P_k - P_k^-.$$

Since (P_k, Q_k) is a strict record gap, $\Delta_k < D_k$. Define

$$y'_k = P_k - \frac{D_k + \Delta_k}{2}.$$

Then $y'_k < P_k^- < P_k < y'_k + D_k < Q_k$. Therefore the interval $(y'_k, y'_k + D_k]$ contains at least the two primes P_k^- and P_k , so

$$\pi(y'_k + D_k) - \pi(y'_k) \geq 2.$$

Also $y'_k \sim P_k$, and (3) gives

$$\frac{D_k}{\log y'_k} \longrightarrow 1.$$

Thus the normalized ratio for $C = 1$ is eventually at least close to 2, not 1, again contradicting $A(1)$. $\square$

4 The difference observation

The following observation is due to Terry Tao. It is the mechanism that turns the preceding obstruction into a structural statement about all constants.

Lemma 1 (Difference closure). *If $A(C_1)$ and $A(C_2)$ hold, then $A(C_1 - C_2)$ holds. Consequently $\mathcal{C}$ is closed under subtraction.*

Proof. We give the argument for $0 < C_2 < C_1$, which is the case needed for positive constants; the same translation argument gives the signed version in the formulation (1). Put

$$\delta = C_1 - C_2, \quad d = d(x).$$

Let $x/2 < y < x$. Since $d = o(x)$, all shifts of y by a fixed multiple of d have logarithm $\log y + o(\log y)$, uniformly in $x/2 < y < x$.

If $y + \delta d < x$, then both y and $y + \delta d$ are admissible starting points. Therefore

$$\begin{aligned} \pi(y + \delta d) - \pi(y) &= [\pi(y + C_1 d) - \pi(y)] - [\pi(y + C_1 d) - \pi(y + \delta d)] \\ &\sim \frac{C_1 d}{\log y} - \frac{C_2 d}{\log(y + \delta d)} \sim \frac{\delta d}{\log y}. \end{aligned}$$

If $y + \delta d \geq x$, put $z = y - C_2 d$. For all large x ,

$$z > x - C_1 d > x/2 \quad \text{and} \quad z < x,$$

so z is admissible. Then

$$\begin{aligned} \pi(y + \delta d) - \pi(y) &= [\pi(z + C_1 d) - \pi(z)] - [\pi(z + C_2 d) - \pi(z)] \\ &\sim \frac{C_1 d}{\log z} - \frac{C_2 d}{\log z} \sim \frac{\delta d}{\log y}. \end{aligned}$$

The error terms are uniform in y , so $A(\delta)$ follows. $\square$

Remark 1. *The displayed proof shows the positive-difference case directly in the original window $x/2 < y < x$. If one works only with positive constants, it proves that the positive admissible constants are closed under positive differences. With the formal two-sided convention for $\mathcal{C}$, Lemma 1 says exactly that $\mathcal{C}$ is an additive subgroup of $\mathbb{R}$.*

5 Structure of the set of constants

Theorem 3. *With $\mathcal{C}$ defined above, either*

$$\mathcal{C} = \{0\},$$

or there is a unique constant $C_0 > 1$ such that

$$\mathcal{C} = C_0 \mathbb{Z} = \{C_0 k : k \in \mathbb{Z}\}.$$

Proof. By Lemma 1, $\mathcal{C}$ is an additive subgroup of $\mathbb{R}$. By Theorem 2,

$$\mathcal{C} \cap (0, 1] = \emptyset.$$

If $\mathcal{C} = \{0\}$, there is nothing to prove. Otherwise define

$$C_0 = \inf(\mathcal{C} \cap (0, \infty)).$$

The preceding exclusion gives $C_0 \geq 1$.

We first show that $C_0 \in \mathcal{C}$. If not, then there are two distinct positive constants $a, b \in \mathcal{C}$ with

$$C_0 < a < b < C_0 + 1.$$

Since $\mathcal{C}$ is closed under subtraction, $b - a \in \mathcal{C}$. But

$$0 < b - a < 1,$$

contradicting $\mathcal{C} \cap (0, 1] = \emptyset$. Hence $C_0 \in \mathcal{C}$. Since no element of $\mathcal{C}$ lies in $0 < C \leq 1$, this also gives $C_0 > 1$.

Now let $C \in \mathcal{C}$ be positive. Choose an integer $m \geq 0$ and a remainder r with

$$C = mC_0 + r, \quad 0 \leq r < C_0.$$

Because $\mathcal{C}$ is an additive subgroup and $C, C_0 \in \mathcal{C}$, the remainder $r = C - mC_0$ also lies in $\mathcal{C}$. By the minimality of C_0 , the only possibility is $r = 0$. Thus every positive element of $\mathcal{C}$ is an integer multiple of C_0 . Since $\mathcal{C}$ is a subgroup and $C_0 \in \mathcal{C}$, all integer multiples of C_0 lie in $\mathcal{C}$. Therefore $\mathcal{C} = C_0\mathbb{Z}$. Uniqueness is immediate, because C_0 is the least positive element of $\mathcal{C}$. $\square$

6 Conclusion

The original universal-in- C assertion is impossible already by Theorem 1. The strengthened structural formulation is sharper: once one records the elementary failure for $0 < C \leq 1$ and the difference closure of the admissible constants, the set of all constants for which the uniform asymptotic can hold is forced to be either trivial or a discrete lattice with generator strictly larger than 1.